
RCC-Drive: Grounding Vision-Language Models in Formal Spatial Calculus for Autonomous Navigation

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Abstract

Vision-language models (VLMs) are gaining attention in autonomous driving due to their ability to unify visual understanding and language-based reasoning, but their practical use remains constrained by high training and adaptation costs, difficulties in incorporating real-time scene structure, and persistent spatial reasoning errors. These failures often manifest as hallucinated pedestrian positions and incorrect topological assertions about road objects, which are especially problematic in safety-critical settings. We investigate the zero-shot spatial reasoning capabilities of VLMs and examine whether explicitly grounding them with Region Connection Calculus-8 (RCC-8) relations, a formal representation of the eight topological relations that may hold between two regions, improves reliability. To this end, we deterministically derive RCC-8 relations from nuScenes sensor data and map annotations, reducing reliance on implicit spatial inference by the model. We evaluate this approach across four safety-critical object-pair types, including pedestrian and vehicle interactions, in order to assess the value of structured topological grounding for autonomous driving scene understanding.

1 Introduction

Vision-language models are gaining traction in autonomous driving due to their ability to unify visual understanding and language-based reasoning [7, 4]. However, their practical deployment remains constrained by high training and adaptation costs, difficulties in incorporating real-time scene structure, and persistent spatial reasoning errors. These failures often manifest as hallucinated pedestrian positions and incorrect topological assertions about road objects, precisely the kinds of errors that are most dangerous in safety-critical settings [3].

We investigate the zero-shot spatial reasoning capabilities of VLMs and examine whether explicitly grounding them with Region Connection Calculus-8 (RCC-8) [6] relations improves reliability. RCC-8 defines a finite, mutually exclusive set of eight topological relations over spatial regions that are deterministically computable from sensor data. Unlike VLM spatial assertions, RCC-8 relations can be computed, logged, and falsified. Prior work in symbolic AI has established that large language models fail to reliably infer RCC-8 relations even when presented with explicit polygon descriptions [2], making a grounding-injection strategy, rather than in-context reasoning, the appropriate design choice.

To this end, we deterministically derive RCC-8 relations directly from nuScenes sensor data and map annotations [1], reducing reliance on implicit spatial inference by the model. The relation between a pedestrian and a crosswalk is a safety-critical, legally defined topological fact with a ground-truth answer derivable from sensor data, and precisely the kind of relation most susceptible to VLM hallucination under camera perspective distortion and partial occlusion.

We present **RCC-Drive**, a training-free system that injects deterministically computed RCC-8 topological constraints into VLM prompts to assess the value of structured topological grounding

for autonomous driving scene understanding. We evaluate across four safety-critical object-pair types, ego-to-pedestrian, pedestrian-to-crosswalk, ego-to-vehicle, and vehicle-to-vehicle, comparing zero-shot VLM performance with and without RCC-8 grounding. Our contributions are:

- A spatial extraction pipeline that converts nuScenes sensor data and map annotations into structured RCC-8 fact sheets over four safety-critical object-pair types, operating entirely on the bird’s-eye-view ground plane in global coordinates (Section 3.2).
- A camera-visibility curation procedure that selects scenes where safety-critical pedestrians are within the calibrated CAM_FRONT field of view, isolating RCC-8 grounding as the sole experimental variable (Section 3.3).
- A prompt construction framework that injects verified RCC-8 fact sheets into VLM inputs under both baseline and grounded conditions, enabling a controlled zero-shot chain of thought setting.
- We further conduct an empirical analysis of risk–accuracy trade-offs and model perception of RCC-8 relations, examining how formal topological grounding affects both predicted risk levels and downstream driving decisions.

2 Related Work

2.1 VLM Spatial Reasoning in Autonomous Driving

Vision-language models have been rapidly integrated into autonomous driving pipelines, yet their spatial reasoning capabilities remain a critical bottleneck. DriveVLM [7] applies chain-of-thought prompting for scene description, analysis, and hierarchical planning, but acknowledges fundamental VLM limitations in spatial understanding and introduces a dual system that falls back on a traditional 3D perception pipeline for robust spatial grounding. SpaceDrive [4] addresses the spatial deficit by replacing textual digit tokens with explicit 3D positional encodings derived from multi-view depth estimation, achieving state-of-the-art open-loop planning on nuScenes [1]. Both approaches rely on Cartesian coordinate representations, bounding boxes or continuous positional embeddings, and require model training or architectural modifications to inject spatial awareness. The SURDS benchmark [3] provides the most direct quantification of this gap: across 9,250 driving-scenario spatial questions built on nuScenes, GPT-4o scores only **13.30/100**, confirming that leading VLMs cannot reliably perform even basic spatial reasoning tasks such as depth estimation, lateral ordering, and front-behind classification in driving contexts. Unlike the above approaches, our method requires no training and no architectural changes; instead, we ground spatial relations in a formal calculus and inject them directly into VLM prompts.

2.2 Region Connection Calculus

The Region Connection Calculus (RCC) was introduced by Randell et al. [6] as a first-order theory of qualitative spatial reasoning built from a single primitive binary relation of *connection*. The RCC-8 fragment defines eight jointly exhaustive and pairwise disjoint (JEPD) topological relations between spatial regions: disconnected (DC), externally connected (EC), partially overlapping (PO), equal (EQ), tangential proper part (TPP) and its inverse (TPPi), and non-tangential proper part (NTPP) and its inverse (NTPPi). Because every pair of regions satisfies exactly one relation, RCC-8 provides **unambiguous, deterministic spatial assertions**, a property we exploit by computing relations from sensor-derived polygons on the bird’s-eye-view ground plane using computational geometry, yielding verified topological facts that require no learned inference.

2.3 LLM Limitations on Formal Spatial Calculi

Cohn and Blackwell [2] provide the definitive evaluation of LLM performance on RCC-8 reasoning. They tested GPT-4, GPT-3.5, Claude-3.5 Sonnet, and Gemini across three tasks, composition table reconstruction, alignment with human composition preferences, and conceptual neighbourhood reconstruction, each with both standard and anonymized relation names, repeating every query 30 times. **No model achieved reliable performance**; accuracy was marginally above chance across all tasks. Models struggled especially with inverse relations (e.g., TPP/TPPi) and performed worse with anonymized names, indicating reliance on memorized associations rather than genuine spatial

reasoning. These findings provide the core motivation for our approach: since LLMs cannot reliably compose or infer RCC-8 relations, we *compute* them deterministically from sensor data and supply them as hard constraints rather than asking the model to derive them.

2.4 Neuro-Symbolic Spatial Grounding

Two recent works explore neuro-symbolic strategies for spatial reasoning but differ from our approach in important respects. Wang et al. [8] propose a DSPy-based pipeline that couples LLMs with Answer Set Programming (ASP) solvers via an iterative feedback loop: the LLM parses natural-language spatial descriptions into ASP facts using RCC-8 relations, and the solver performs deductive reasoning. This achieves 40–50% accuracy gains on text-based spatial benchmarks, but operates exclusively on textual input with no visual perception or driving application. Wu et al. [9] introduce Constraint-Aware Prompting (CAP), which injects bidirectional and transitivity constraints into VLM prompts to reduce spatial hallucinations, a training-free, prompt-level approach closest in spirit to ours. However, CAP operates on generic images with informal spatial language (e.g., “left of,” “above”), lacks a formal spatial calculus, and includes no safety-critical evaluation. RCC-Drive bridges these gaps: it combines the formal rigor of RCC-8 with the prompt-injection strategy of CAP, grounds relations in verified sensor geometry rather than text parsing, and targets the safety-critical autonomous driving domain on nuScenes.

3 Method

3.1 Dataset

We use the nuScenes v1.0-mini split [1] for development and validation, with yield estimates on v1.0-trainval for scale. nuScenes records synchronized sensor data, camera, LiDAR, RADAR, and GPS/IMU, from a Renault Zoé instrumented vehicle across urban scenes in Boston and Singapore. Each *sample* is a synchronized keyframe at approximately 2 Hz. Per-sample 3D bounding box annotations for all traffic participants are provided, and the separately distributed Map Expansion v1.3 supplies `ped_crossing` vector polygon layers in globally registered coordinates.

3.2 Spatial Fact Extraction

3.2.1 Coordinate Frame

All RCC-8 relations are computed on the **bird’s-eye-view (BEV) ground plane in nuScenes global coordinates**. This is a deliberate methodological choice driven by three constraints. First, the ego vehicle has no corresponding camera mask and therefore cannot participate in any image-plane RCC-8 relation. Second, crosswalks are stored as vector polygons in global coordinates in the map layer with no image-space annotation. Third, camera perspective projection introduces physically misleading topology: a pedestrian occluded behind a vehicle overlaps it in image space but is spatially disjoint on the ground plane. Since the core safety question, “is the pedestrian currently on the crosswalk?”, is a ground-plane question, BEV global coordinates are the semantically correct frame.

3.2.2 Region Extraction

Each spatial participant is represented as a Shapely polygon in the global BEV plane.

Ego vehicle. A rotated rectangle of 4.084 m $\times$ 1.730 m (Renault Zoé factory dimensions), centered at the ego pose translation, rotated by the yaw extracted from the pose quaternion. The ego pose is retrieved via the `LIDAR_TOP sample_data` record, which carries a globally registered ego pose at each keyframe.

Traffic participants. `sample_annotation` records in nuScenes store bounding box translation and rotation directly in global coordinates, no sensor-to-ego transform is required. Each annotation’s footprint is a rotated rectangle from its `size` field and yaw.

Crosswalks. Queried from the Map Expansion `ped_crossing` layer using an 80 m search radius from ego via the nuScenes `NuScenesMap.get_records_in_radius` API.

All footprints are repaired with `buffer(0)` before predicate evaluation to resolve invalid geometries arising from near-collinear annotation data.

3.2.3 RCC-8 Classifier

We implement the full eight-relation RCC-8 calculus [6] using Shapely’s DE-9IM predicates as a decision tree:

1. **Degenerate area guard.** $\text{Area} < 10^{-8} \text{ m}^2 \Rightarrow \text{DC}$.
2. `equals_exact( $\epsilon = 10^{-6}$ )` $\Rightarrow \text{EQ}$
3. `disjoint()` $\Rightarrow \text{DC}$
4. `touches()` $\Rightarrow \text{EC}$
5. `a.within(b)` $\Rightarrow \text{a.boundary.intersects(b.boundary)} \Rightarrow \text{TPP (true) or NTPP (false)}$
6. `b.within(a)` $\Rightarrow \text{b.boundary.intersects(a.boundary)} \Rightarrow \text{TPP}^{-1} \text{ (true) or NTPP}^{-1} \text{ (false)}$
7. Otherwise $\Rightarrow \text{PO}$

An important implementation note: Shapely’s `within()` predicate returns `True` for both TPP and NTPP configurations and is therefore insufficient for distinguishing them. The additional `boundary.intersects()` check is required; this behavior was empirically verified and is covered by unit tests for all eight relations.

In practice, only DC, NTPP, and PO appear across the 12,834 relations computed from v1.0-mini. EC, EQ, TPP, TPP^{-1} , and NTPP^{-1} do not occur in real sensor data, which is physically expected: GPS/sensor-fused BEV localization has enough position noise that exact boundary contact between independent objects never manifests.

3.2.4 Pair Types and Filtering

We compute RCC-8 for four pair types, each selected for a distinct safety rationale:

Table 1: Object-pair types computed by the extraction pipeline.

Pair Type	Safety Rationale
<code>ego_vs_pedestrian</code>	Direct collision risk
<code>pedestrian_vs_crosswalk</code>	Legal crossing context
<code>ego_vs_vehicle</code>	Direct collision risk
<code>vehicle_vs_vehicle</code>	Surrogate situational awareness

Annotations farther than 50 m from ego are excluded before computation. Vehicle-vehicle pairs undergo two-stage filtering: pairs whose centroids are more than 20 m apart are never classified (distant vehicle topology is spatially uninformative), and among the remaining pairs, DC relations are additionally dropped. DC pairs for pedestrian-crosswalk are likewise dropped. Across v1.0-mini, 53,167 vehicle-vehicle DC pairs were suppressed by these filters, validating the design: without filtering, DC would dominate the relation distribution and dilute the actionable signal injected into prompts. Drop counts are preserved in each output file’s `metadata` field for auditability.

3.3 Camera Visibility Curation

Phase 2 evaluation is restricted to scenes where at least one *safety-critical pedestrian*, defined as a pedestrian with an NTPP or PO relation to a crosswalk, falls within the `CAM_FRONT` field of view. This restriction ensures the baseline VLM has visual access to the relevant object, making RCC-8 grounding the sole experimental variable rather than confounding VLM failure with visual invisibility.

The `CAM_FRONT` half-FOV is computed per sample from the calibrated sensor intrinsic matrix:

$$\theta_{\text{half}} = \arctan\left(\frac{w}{2f_x}\right) \quad (1)$$

where $f_x = K_{00}$ and w is the image width from the `sample_data` record. Across all 404 v1.0-mini samples, the measured half-FOV spans 32.28° – 32.56° (mean 32.42°), a variance of less than 0.3° , confirming sensor hardware consistency across logs.

3.4 Dataset Yield

Table 2: Curated sample counts after spatial extraction and FOV filtering. Trainval count is an estimate from metadata without full extraction.

Split	Total	Peds + Crosswalks	Curated	Yield
v1.0-mini	404	285	59	14.6%
v1.0-trainval (est.)	34,149	—	~3,398	~10.0%

The yield difference between splits is expected: mini scenes were hand-selected for dense urban environments, while trainval includes quiet roads and highway segments.

4 Experiments

4.1 Experimental Setup

We evaluate **RCC-Drive** on the curated safety-critical subset of *nuScenes v1.0-mini* described in Section 3.3. This subset contains 59 keyframes selected to ensure that at least one pedestrian relevant to crossing behavior is visible in the front camera view and situated in a scene where crosswalk topology is meaningful. Our experiments are designed to test whether injecting formally computed topological relations changes how a vision-language model interprets driving scenes, assigns risk, and selects actions.

We use the same vision-language model, *MiniCPM-V*, for all conditions, with temperature 0.0. The image input is always the CAM_FRONT frame. The only difference between conditions is whether the prompt includes the structured RCC-8 fact sheet derived from LiDAR and HD map geometry.

4.2 Prompting Conditions

We compare two prompting conditions.

Baseline prompt. The model receives only the front-camera image together with a structured driving prompt asking it to describe the scene, reason about the spatial layout, assess the risk level, and choose a driving action. This is inspired by the chain-of-thought approach in prior literature [5].

RCC-Drive prompt. The model receives the same image together with a structured list of RCC-8 spatial relations computed from sensor and map data. These relations provide explicit topological grounding over safety-relevant object pairs such as ego–pedestrian, pedestrian–crosswalk, ego–vehicle, and vehicle–vehicle.

Both prompts are aligned in format and output schema so that the presence of topological facts is the primary experimental variable. Although only a subset of RCC-8 relations appears in the current nuScenes-based extraction pipeline, we retain the full RCC-8 relation vocabulary in the prompt definition for two reasons. First, keeping the full schema makes the prompt reusable across other driving datasets and deployment settings where additional RCC-8 relations may arise. Second, exposing the model to the complete formal relation set allows us to probe whether VLMs interpret topological relations in driving scenes in ways that align with, or diverge from, human understanding of those relations.

Baseline Chain of thought Prompt

You are an autonomous driving scene analyst.

You are given a front-camera image from a vehicle (nuScenes dataset).

Describe the driving scene and decide what the vehicle should do.

1. **Scene Description:** What objects do you see? Describe briefly.

2. **Spatial Reasoning:** Describe the general layout of the scene.

3. **Risk Assessment:** Rate the overall risk as exactly one of: LOW, MEDIUM, HIGH, CRITICAL.

4. **Driving Decision:** Choose exactly one action:

MAINTAIN_SPEED, DECELERATE, BRAKE, EMERGENCY_STOP,
LANE_CHANGE_LEFT, LANE_CHANGE_RIGHT

Respond ONLY with valid JSON (no other text):

```
{  
  "scene_description": "<description>",  
  "spatial_reasoning": "<reasoning>",  
  "risk_level": "<LOW|MEDIUM|HIGH|CRITICAL>",  
  "risk_explanation": "<explanation>",  
  "driving_decision": "<action>",  
  "decision_justification": "<justification>"  
}
```

RCC-DRIVE Chain of Thought Prompt

You are an autonomous driving scene analyst.

You are given:

1. A front-camera image from a vehicle in the nuScenes dataset.
2. Spatial relations between objects in the scene, computed from LiDAR and HD map data using RCC-8 topology. These relations represent measured geometric relationships between object regions.

RCC-8 Relation Definitions

- **DC** (Disconnected): The regions do not touch; a measured gap exists between them.
- **EC** (Externally Connected): The region boundaries touch; the gap is zero.
- **PO** (Partial Overlap): The regions intersect; one partially occupies space within the other.
- **TPP** (Tangential Proper Part): One region lies inside another and touches its boundary.
- **NTPP** (Non-Tangential Proper Part): One region lies entirely inside another without touching its boundary.

Measured Spatial Relations for This Scene

Ego Vehicle vs. Pedestrians

- Ego is disconnected from adult.
- Ego is externally connected to adult.

Pedestrians vs. Crosswalks

- adult is a non-tangential proper part of crosswalk.
- adult partially overlaps with crosswalk.
- child is disconnected from crosswalk.

Ego Vehicle vs. Other Vehicles

- Ego is disconnected from car.
- Ego is externally connected to truck.

Vehicle vs. Vehicle

- car is externally connected to truck.
- car partially overlaps with car.

Analyze the driving scene using both the image and the measured spatial relations.

1. **Scene Description:** Briefly describe the visible objects in the scene and cross-reference them with the provided spatial relations.
2. **Spatial Reasoning:** Use the RCC-8 relations to describe the scene layout precisely. Pay particular attention to EC, PO, TPP, and NTPP relations, as these indicate physical contact, overlap, or close containment. Cite the relevant relations explicitly.
3. **Risk Assessment:** Assign exactly one risk level: LOW, MEDIUM, HIGH, or CRITICAL. Your assessment must account for the measured spatial relations, especially cases involving EC, PO, TPP, or NTPP.
4. **Driving Decision:** Choose exactly one action from:
MAINTAIN_SPEED, DECELERATE, BRAKE, EMERGENCY_STOP, LANE_CHANGE_LEFT, LANE_CHANGE_RIGHT

Respond only with valid JSON:

```
{
  "scene_description": "<description>",
  "spatial_reasoning": "<reasoning citing RCC-8 relations>",
  "risk_level": "<LOW|MEDIUM|HIGH|CRITICAL>",
  "risk_explanation": "<explanation>",
  "driving_decision": "<action>",
  "decision_justification": "<justification>"
}
```

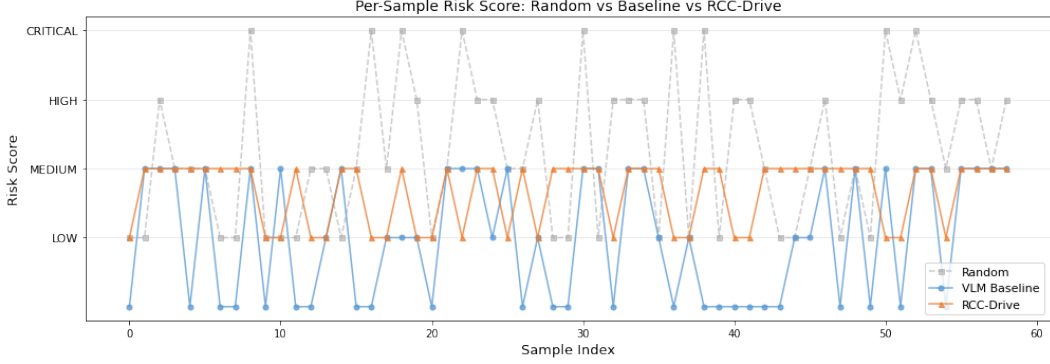


Figure 1: Risk predictions across the safety-critical subset for the random baseline, image-only VLM baseline, and RCC-Drive. The figure shows that RCC-Drive often assigns different risk levels from the image-only baseline on the same scenes, indicating that explicit RCC-8 spatial relations influence perceived scene severity.

4.3 Evaluation Protocol

Our current evaluation focuses on **qualitative risk analysis**. The goal is not yet to claim full end-task driving performance, but to study how explicit topological grounding changes model behavior on the same scene. For each sample, the model outputs one of four risk levels—LOW, MEDIUM, HIGH, or CRITICAL—and one driving action from a fixed set of maneuvers.

We analyze the outputs along three dimensions. First, we compare the overall distribution of predicted risk levels under the Baseline and RCC-Drive conditions. Second, we compare the distribution of selected driving actions. Third, we inspect scene-level differences between the two conditions to determine whether adding RCC-8 facts shifts the model toward more cautious, less cautious, or simply different interpretations of the same visual input. To contextualize these predictions, we also report a random reference baseline that uniformly samples from the four risk labels.

This protocol is intended as a first step toward understanding whether deterministic topological grounding materially influences VLM scene interpretation in safety-critical driving settings.

4.4 Risk Distribution Analysis

Figure 1 shows the risk labels predicted across the curated safety-critical subset for the random baseline, the image-only baseline prompt, and RCC-Drive. As expected, the random baseline exhibits noisy label variation with no consistent structure. In contrast, both prompt-based model conditions show more stable and scene-dependent behavior.

Most importantly, RCC-Drive frequently assigns different risk labels from the image-only baseline on the same scenes. This indicates that the injected RCC-8 facts are not being ignored: they alter how the model interprets scene severity. At this stage, we interpret these shifts as evidence that explicit topological grounding changes the model’s internal assessment of safety-relevant situations, especially in scenes involving pedestrians, crosswalk occupancy, and nearby vehicles.

These results should be understood as an analysis of *risk perception under grounding*, rather than as a definitive measure of downstream driving correctness. The present experiments establish that formal spatial facts affect model outputs; subsequent analyses are needed to determine when these changes correspond to improved calibration, stronger safety alignment, or more defensible reasoning.

4.5 Planned Follow-Up Analyses

We plan three follow-up analyses: (1) a risk transition matrix to measure how RCC-Drive shifts risk labels relative to the baseline on the same scenes, (2) risk–decision heatmaps to test whether selected actions align more consistently with assigned risk, and (3) qualitative comparisons of low-risk and high-risk cases to inspect whether RCC-8 facts lead to more spatially precise reasoning. Together,

these analyses will help determine whether topological grounding improves the consistency and interpretability of model behavior in safety-critical driving scenes.

4.6 Quantitative Safety Analysis: Accuracy and Risk Trade-off

We quantify the effect of RCC-8 grounding through an accuracy–risk trade-off analysis across the 59 curated scenes. For each scene, the model produces a driving decision under two conditions: the ungrounded baseline and the RCC-8-augmented prompt. We measure:

Decision Accuracy. Whether the model’s generated decision matches the ground-truth action. This is computed as the fraction of scenes in which the model’s output is consistent with the ground-truth relational configuration.

Risk Score. Each generated decision is assigned a risk level based on a standardized rubric that maps action–topology pairs to risk categories. The key quantity of interest is the *accuracy–risk trade-off*: whether a decision that is accurate (matches ground truth) also correctly captures the risk level implied by the topology. For instance, a model may correctly select “brake” in a scene where a pedestrian is NTPP with a crosswalk, but incorrectly assign low risk to a scene where the same action is warranted by a PO relation, indicating that the model is right for the wrong reasons.

This trade-off analysis is intended to identify cases where RCC-8 grounding improves decision accuracy but does not fully resolve the model’s underlying risk miscalibration, as well as cases where the grounded model is accurate and risk-calibrated, providing evidence that topological constraints are sufficient to guide safe decision-making.

5 Conclusion

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